

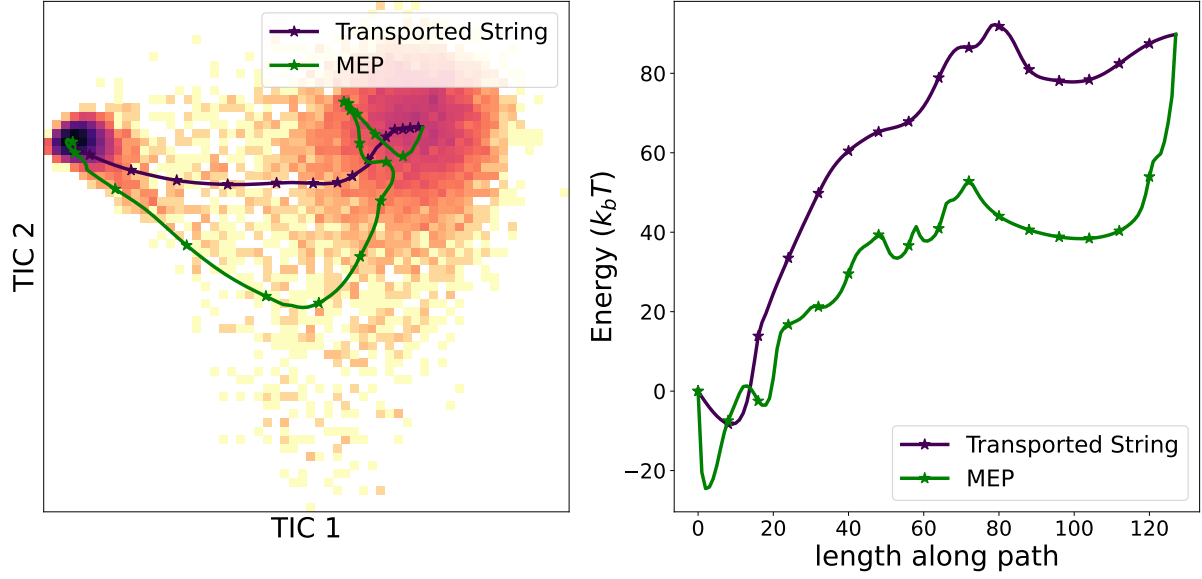


Figure 1: **Projection on TICA and free energy profile of Chignolin.** Projection on the first 2 TIC coordinates (left) and free energy computed as logarithm of the probability for a folding trajectory of the protein Chignolin. The transported string is the string obtained by transport alone ($\gamma = 0$). The MEP move toward regions of lower free energy on the TICA landscape, as they should.

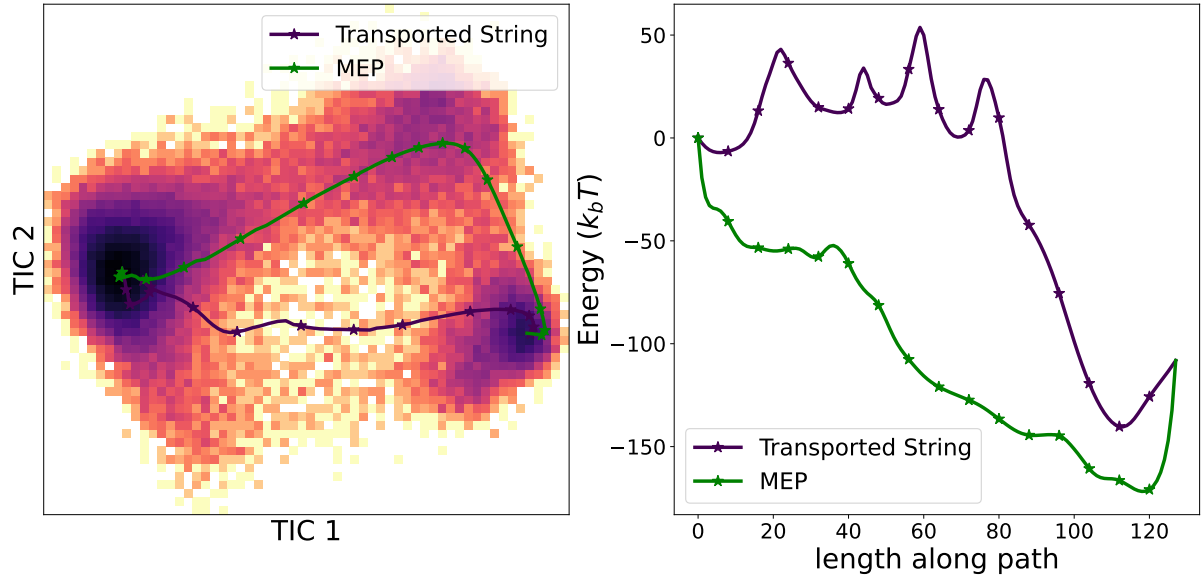


Figure 2: **Projection on TICA and free energy profile of BBA.** Projection on the first 2 TIC coordinates (left) and free energy computed as logarithm of the probability for a folding trajectory of the protein BBA. The transported string is the string obtained by transport alone ($\gamma = 0$). The MEP move toward regions of lower free energy on the TICA landscape, as they should.

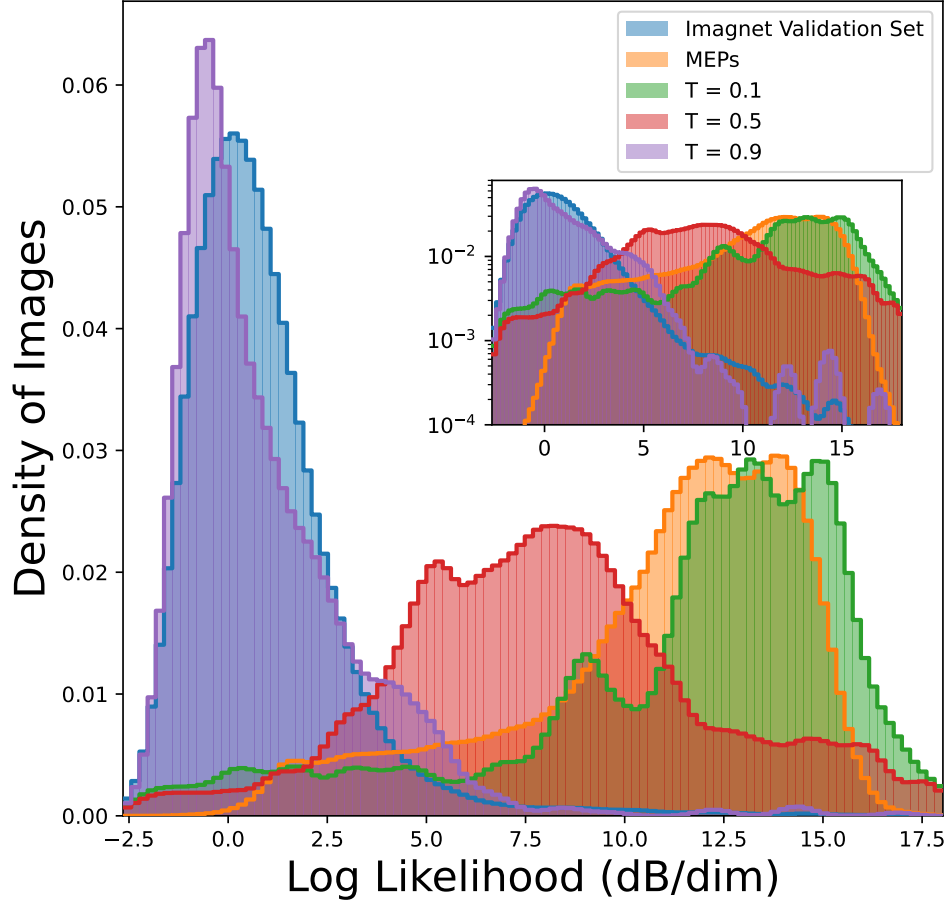


Figure 3: **Histograms of the likelihood for the different methods.** Histogram of the likelihoods of the images for strings obtained using the different methods compared to the histogram of the likelihood of the validation set of Imagnet. The images along the principal curve have likelihood similar to the images in the validation set, indicative of their typicality. The images along the MEP have much higher likelihood, and are atypical. As the temperature is increased along the string, the likelihood of images decrease in a way that make them move toward the typical set, as expected.

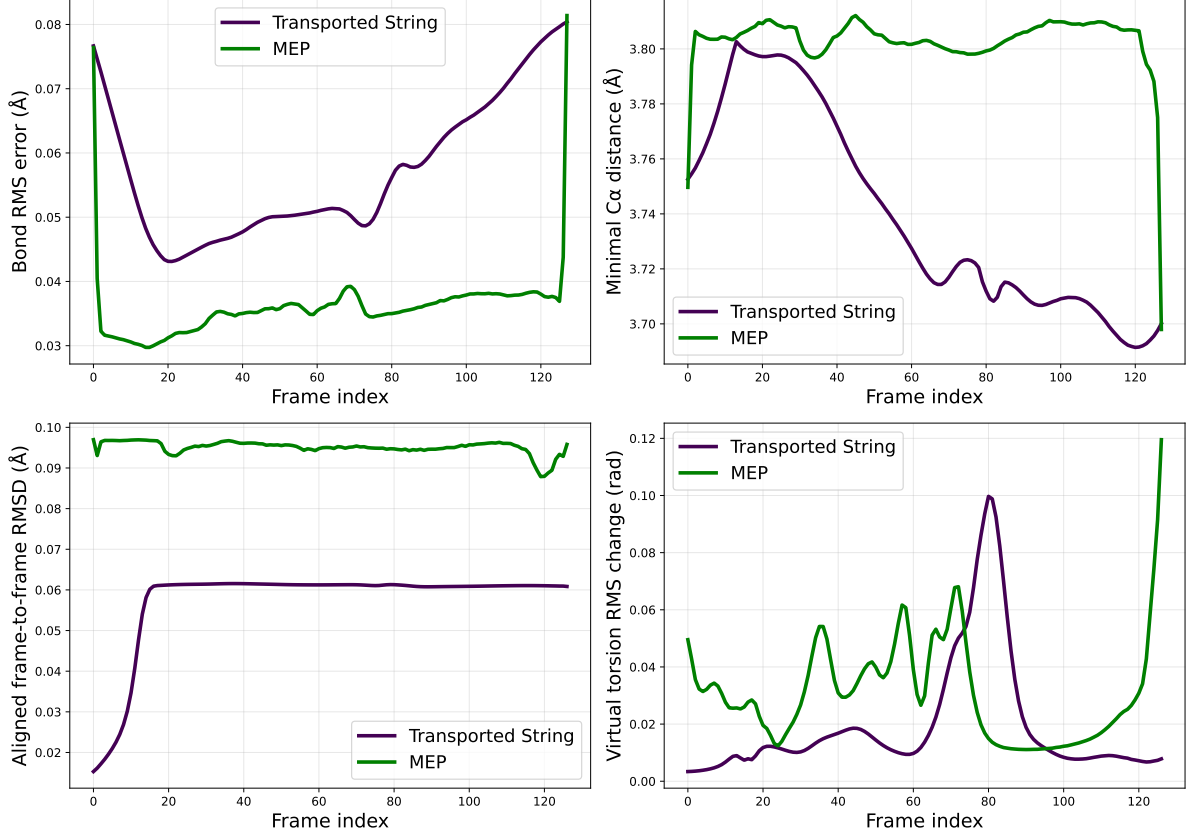


Figure 4: **Realism and smoothness metrics for the Chignolin Trajectories in Fig.1** (Upper left) RMS error of consecutive $C\alpha$ distances relative to the canonical 3.8 Å; the MEP remains closer to this reference throughout. (Upper right) Minimum pairwise $C\alpha$ distance; both strings stay above 3.7 Å, making backbone steric clashes unlikely. (Lower left) Frame-to-frame RMSD, measuring trajectory smoothness; the MEP is markedly more uniform, as enforced by the string method. (Lower right) RMS variation of dihedral angles between consecutive frames; this remains small across all frames for the MEP.

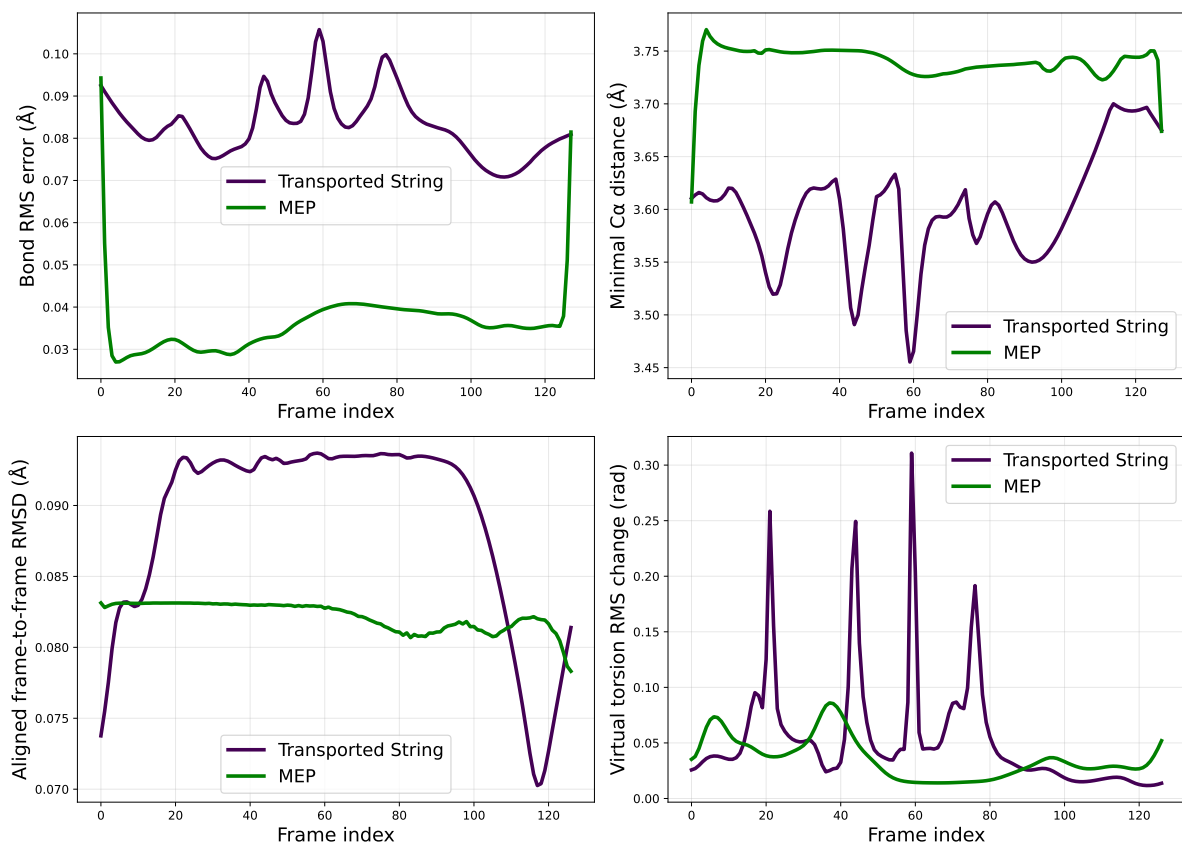


Figure 5: **Realism and smoothness metrics for the BBA Trajectories in Fig.2** Same metrics as Fig. 4 for the BBA trajectories. The MEP again shows better backbone geometry and smoother progression. Notably, the transported string reaches minimum $C\alpha$ distances as low as 3.47 Å, making steric clashes more likely, while the MEP stays consistently above 3.7 Å.

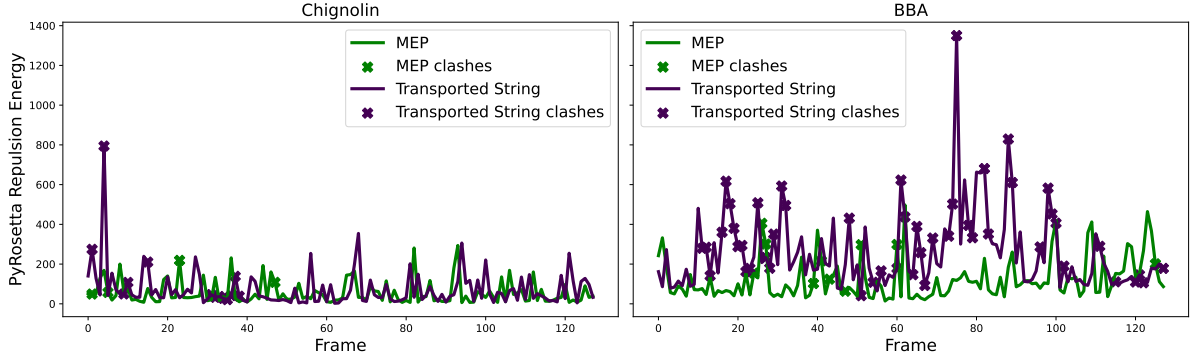


Figure 6: **Reconstructed Repulsion Energy and Steric Clashes for the trajectories in Fig.1 (Chignolin) and Fig.2 (BBA)** PyRosetta repulsive energies and side-chain steric clashes for Chignolin (left) and BBA (right). Since diffusion models output backbone coordinates only, all-atom structures were reconstructed with PULCHRA and side chains relaxed with PyRosetta (backbone fixed). Frames with detected side-chain clashes are marked with $\times$. This evaluation is model-independent: the force field was not used during training. The MEP consistently achieves lower repulsive energies and substantially fewer clashes than the transported string, providing external physical validation of the pathway quality.